Practice Problems: Linear Algebra

ECON 441: Introduction to Mathematical Economics — Instructor: Div Bhagia

Exercise 4.2

1. Given $A = \begin{bmatrix} 7 & -1 \\ 6 & 9 \end{bmatrix}$, $B = \begin{bmatrix} 0 & 4 \\ 3 & -2 \end{bmatrix}$, and $C = \begin{bmatrix} 8 & 3 \\ 6 & 1 \end{bmatrix}$, find:
   (a) $A + B$
   (b) $C - A$
   (c) $3A$
   (d) $4B + 2C$

2. Given $A = \begin{bmatrix} 2 & 8 \\ 3 & 0 \\ 5 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 2 & 0 \\ 3 & 8 \end{bmatrix}$, and $C = \begin{bmatrix} 7 & 2 \\ 6 & 3 \end{bmatrix}$:
   (a) Is AB defined? Calculate AB. Can you calculate BA? Why?
   (b) Is BC defined? Calculate BC. Is CB defined? If so, calculate CB. Is it true that $BC = CB$.

4. Find the product matrices in the following (in each case, append beneath every matrix a dimension indicator):
   (a) $\begin{bmatrix} 0 & 2 & 0 \\ 3 & 0 & 4 \\ 2 & 3 & 0 \end{bmatrix} \begin{bmatrix} 8 & 0 \\ 0 & 1 \\ 3 & 5 \end{bmatrix}$
   (b) $\begin{bmatrix} 6 & 5 & -1 \\ 1 & 0 & 4 \end{bmatrix} \begin{bmatrix} 4 & -1 \\ 5 & 2 \\ 0 & 1 \end{bmatrix}$
   (c) $\begin{bmatrix} 3 & 5 & 0 \\ 4 & 2 & -7 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$
   (d) $\begin{bmatrix} a & b & c \end{bmatrix} \begin{bmatrix} 7 & 0 \\ 0 & 2 \\ 1 & 4 \end{bmatrix}$

Exercise 4.4

5. (e) Find (i) $C = AB$, and (ii) $D = BA$, if

$$A = \begin{bmatrix} -2 \\ 4 \\ 7 \end{bmatrix} \quad B = \begin{bmatrix} 3 & 6 & -2 \end{bmatrix}$$

7. If the matrix A in Example 5 had all its four elements nonzero, would $x^T Ax$ still give a weighted sum of squares? Would the associative law still apply?

Exercise 4.5

1. Given $A = \begin{bmatrix} -1 & 5 & 7 \\ 0 & -2 & 4 \end{bmatrix}$, $b = \begin{bmatrix} 9 \\ 6 \\ 0 \end{bmatrix}$, and $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$:

   Calculate: (a) AI (b) IA (c) Ix (d) $x^T I$
   Indicate the dimension of the identity matrix used in each case.

4. Show that the diagonal matrix

$$\begin{bmatrix} a_{11} & 0 & \cdots & 0 \\ 0 & a_{22} & \cdots & 0 \\ \cdots & \cdots & \cdots & \cdots \\ 0 & 0 & \cdots & a_{nn} \end{bmatrix}$$

   can be idempotent only if each diagonal element is either 1 or 0 . How many different numerical idempotent diagonal matrices of dimension $n \times n$ can be constructed altogether from such a matrix?

Exercise 4.6

2. Given $A = \begin{bmatrix} 0 & 4 \\ -1 & 3 \end{bmatrix}$, $B = \begin{bmatrix} 3 & -8 \\ 0 & 1 \end{bmatrix}$, and $C = \begin{bmatrix} 1 & 0 & 9 \\ 6 & 1 & 1 \end{bmatrix}$, verify that

   (a) $(A + B)^T = A^T + B^T$

   (b) $(AC)^T = C^T A^T$

6. Let $A = I - X\left(X^T X\right)^{-1} X^T$.

   (a) Must A be square? Must $\left(X^T X\right)$ be square? Must X be square?

   (b) Show that matrix A is idempotent.

Exercise 5.1

3. Are the rows linearly independent in each of the following?

(a) $\begin{bmatrix} 24 & 8 \\ 9 & -3 \end{bmatrix}$ (c) $\begin{bmatrix} 0 & 4 \\ 3 & 2 \end{bmatrix}$

(b) $\begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$ (d) $\begin{bmatrix} -1 & 5 \\ 2 & -10 \end{bmatrix}$

4. Check whether the columns of each matrix in Prob. 3 are also linearly independent. Do you get the same answer as for row independence?

Exercise 5.2

1. Evaluate the following determinants:

(c) $\begin{vmatrix} 4 & 0 & 2 \\ 6 & 0 & 3 \\ 8 & 2 & 3 \end{vmatrix}$ (e) $\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}$ (f) $\begin{vmatrix} x & 5 & 0 \\ 3 & y & 2 \\ 9 & -1 & 8 \end{vmatrix}$

2. Determine the signs to be attached to the relevant minors in order to get the following cofactors of a determinant: $|C_{13}|, |C_{23}|, |C_{33}|, |C_{41}|$, and $|C_{34}|$.

3. Given $\begin{vmatrix} a & b & c \\ d & e & f \\ g & b & i \end{vmatrix}$, find the minors and cofactors of the elements a, b and f.

6. Find the minors and cofactors of the third row, given

$$A = \begin{bmatrix} 9 & 11 & 4 \\ 3 & 2 & 7 \\ 6 & 10 & 4 \end{bmatrix}$$

Exercise 5.3

1. Use the determinant $\begin{vmatrix} 4 & 0 & -1 \\ 2 & 1 & -7 \\ 3 & 3 & 9 \end{vmatrix}$ to verify the first four properties of determinants.

4. Show that when all the elements of an nth-order determinant |A| are multiplied by a number k, the result will be $k^n|A|$.

5. Calculate the determinant for the following matrices. Comment on whether the matrices are nonsingular and the rank of each matrix.

(a) $\begin{bmatrix} 4 & 0 & 1 \\ 19 & 1 & -3 \\ 7 & 1 & 0 \end{bmatrix}$ (c) $\begin{bmatrix} 7 & -1 & 0 \\ 1 & 1 & 4 \\ 13 & -3 & -4 \end{bmatrix}$

(b) $\begin{bmatrix} 4 & -2 & 1 \\ -5 & 6 & 0 \\ 7 & 0 & 3 \end{bmatrix}$ (d) $\begin{bmatrix} -4 & 9 & 5 \\ 3 & 0 & 1 \\ 10 & 8 & 6 \end{bmatrix}$

8. Comment on the validity of the following statements:

   (a) Given any matrix A, we can always derive from it a transpose and a determinant.

   (b) Multiplying each element of an $n \times n$ determinant by 2 will double the value of that determinant.

   (c) If a square matrix A vanishes, then we can be sure that the equation system $Ax = d$ is nonsingular.

Exercise 5.4

2. Find the inverse of each of the following matrices:

   (a) $A = \begin{bmatrix} 5 & 2 \\ 0 & 1 \end{bmatrix}$ (c) $C = \begin{bmatrix} 3 & 7 \\ 3 & -1 \end{bmatrix}$

   (b) $B = \begin{bmatrix} -1 & 0 \\ 9 & 2 \end{bmatrix}$ (d) $D = \begin{bmatrix} 7 & 6 \\ 0 & 3 \end{bmatrix}$

3. (a) Drawing on your answers to Prob. 2, formulate a two-step rule for finding the adjoint of a given 2×2 matrix A: In the first step, indicate what should be done to the two diagonal elements of A in order to get the diagonal elements of $adj A$; in the second step, indicate what should be done to the two off-diagonal elements of A. (Warning: This rule applies only to 2×2 matrices.)

   (b) Add a third step which, in conjunction with the previous two steps, yields the 2×2 inverse matrix A^{-1}.

4. Find the inverse of each of the following matrices:

(a) $E = \begin{bmatrix} 4 & -2 & 1 \\ 7 & 3 & 0 \\ 2 & 0 & 1 \end{bmatrix}$

(b) $F = \begin{bmatrix} 1 & -1 & 2 \\ 1 & 0 & 3 \\ 4 & 0 & 2 \end{bmatrix}$

(c) $G = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$

(d) $H = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

7. Is it possible for a matrix to be its own inverse?

Exercise 5.4

6. Solve the system $Ax = d$ by matrix inversion, where

(a) $\begin{aligned} 4x + 3y &= 28 \\ 2x + 5y &= 42 \end{aligned}$

(b) $\begin{aligned} 4x_1 + x_2 - 5x_3 &= 8 \\ -2x_1 + 3x_2 + x_3 &= 12 \\ 3x_1 - x_2 + 4x_3 &= 5 \end{aligned}$

Exercise 5.5

1. Use Cramer's rule to solve the following equation systems:

(a) $\begin{aligned} 3x_1 - 2x_2 &= 6 \\ 2x_1 + x_2 &= 11 \end{aligned}$

(b) $\begin{aligned} -x_1 + 3x_2 &= -3 \\ 4x_1 - x_2 &= 12 \end{aligned}$

(c) $\begin{aligned} 8x_1 - 7x_2 &= 9 \\ x_1 + x_2 &= 3 \end{aligned}$

(d) $\begin{aligned} 5x_1 + 9x_2 &= 14 \\ 7x_1 - 3x_2 &= 4 \end{aligned}$

2. For each of the equation systems in Prob. 1, find the inverse of the coefficient matrix, and get the solution by the formula $x^* = A^{-1}d$.

3. Use Cramer's rule to solve the following equation systems:

(a) $\begin{aligned} 8x_1 - x_2 &= 16 \\ 2x_2 + 5x_3 &= 5 \\ 2x_1 + 3x_3 &= 7 \end{aligned}$

(d) $\begin{aligned} -x + y + z &= a \\ x - y + z &= b \\ x + y - z &= c \end{aligned}$